Bhanu Anish Akkineni

San Antonio, TX | (210) 508-1380 | anish.akkineni9@gmail.com | LinkedIn

EDUCATION

University of Texas at San Antonio – San Antonio, TX
Master of Science in ***Computer Science*** (GPA: 3.96/4.00) August 2023 – May 2025

P V P Siddhartha Institute of Technology – Vijayawada, India
Bachelor of Technology in ***Computer Science*** (GPA: 3.8/4.00) August 2019 – May 2023
Relevant Coursework: Operating systems, RDBMS, Algorithms, OOP, Cloud Computing, AI ML, Web Development, Computer Architecture, Computer Networks.

WORK EXPERIENCE

Computer Science Department, UTSA – San Antonio, TX July 2025 - Present
Research Assistant

- Apply deep learning models using TensorFlow/PyTorch to optimize forecasting and integration in hybrid renewable energy systems.
- Conduct literature reviews and evaluated AI-based optimization methods for multi-source renewable energy modeling and efficiency improvement.
- Support data preprocessing, model training, and performance analysis; collaborate on research publications and reports

Computer Science Department, UTSA – San Antonio, TX August 2023 – May 2025
Grader / Teaching Assistant

- Evaluated assignments and projects in Data Structures and Algorithms, Python programming, Database design and management and Systems Programming, applying foundational software engineering concepts
- Reviewed student code in Python, JavaScript, HTML/CSS, and Unix/Linux shell, ensuring correctness, performance, and maintainability and provided valuable feedback to improve in the topics

SYTIQHUB Educational Services Private Limited – Vijayawada, India February 2023 – August 2023
Software Engineer Intern

- Built and maintained full-stack web application for small business using React (frontend) with Redux and Python FastAPI (backend), developing RESTful APIs and optimizing MySQL queries using Database ORMs like SQLALchemy to improve feature delivery speed by 20% and reduce latency for critical workflows.
- Implemented CI/CD pipelines with GitHub Actions to automate deployments and integrated testing frameworks (PyTest, Jest) to ensure reliability, maintainability, and seamless delivery of new features.
- Collaborated in an Agile environment, contributing to sprint planning, peer code reviews, and building scalable, efficient solutions that enhanced UI responsiveness and supported large-scale data handling.

PUBLICATIONS

A Network Monitoring Model based on CNN for Unbalanced Network Activity (*Link*) November 2022 – March 2023

- Research done as part of the project, results and findings were accepted by ICSSIT 2023 International Conference. A research paper was published in IEEE explore
- Worked on a Deep Learning project to classify network traffic as normal or malicious using the NSL-KDD dataset
- Performed thorough data preprocessing including feature standardization, label encoding and applied Linear Discriminant Analysis to reduce dimensionality and enhance separability of network behavior patterns
- Designed and trained a CNN with the Adam optimizer, achieving ~98% accuracy, 97% F1-score, and ROC-AUC of 0.99 for intrusion detection and interpreted model reliability in real-world scenarios

SKILLS

Programming: Java, Python, C++, SQL
RDBMS: MS SQL server, MySQL
Tools and Technologies: HTML, CSS, JavaScript, SQLALchemy, React, Bootstrap, REST APIs, Git, Unix/Linux, Excel, AWS fundamentals, Google Sheets, Business intelligence tools, NoSQL, CI/CD, Database ORM
Other skills: Unit Testing, LLD, Data Structures and Algorithms, Team Collaboration, Problem Solving, Communication

PROJECTS

Customer and Product Data Warehouse

- Designed and implemented a Modern Data Warehouse using Medallion Architecture (Bronze, Silver, Gold) to streamline data processing and analytics
- Built robust ETL pipelines to ingest raw data from CSV files into SQL Server, performing data cleansing, transformation, and normalization in successive layers
- Modeled data into star schema structures with fact and dimension tables optimized for reporting and analytical queries
- Applied industry best practices in data engineering and analytics to enable business-ready insights and support scalable reporting solutions

Spam Email Detection using Machine Learning model

- Designed and implemented a modular ML pipeline using scikit-learn to classify spam emails based on text data
- Automated preprocessing tasks (cleaning, lemmatization, vectorization) with reusable and maintainable code components
- Integrated feature engineering and Random Forest into a scalable training workflow with robust evaluation
- Achieved 96% accuracy, 94% F1-score, and 0.98 ROC-AUC, demonstrating high model reliability